WACV #*****

EquiOrient Phase 2: Supplementary Material

Anonymous WACV Datasets Track submission

Paper ID *****

1. Full 45-test verification suite

All 45 tests must pass before any GPU run. The suite catches the entire class of bugs discovered in Phase 1 (zero structural losses, wrong label copying, stale data volumes).

Category	Tests
D4 group axioms (4)	identity, $R^4{=}I$, $H^2{=}I$, $HRH{=}R^{-1}$
Label action (3)	direction_H, direction_R, label_composition_all
Renderer (4)	H/R/R2/R3 pixel transform, bbox transform, object identity
Dataset integrity (5)	scene disjoint, balanced labels, no boundary, transform coverage, unique pngs
Structural loss (4)	equivariance nonzero, wrong nonzero, wrong differs, manipulation check
Gradient flow (5)	grad reaches pair_encoder (x2), vision_lora, rho never receives label, common init
Architecture (3)	matched init, equal param count, z_ablation
Collapse checks (4)	variance, effective rank, norm, dead neurons
Identifiability (2)	collision matrix, unseen distinguishes correct/wrong

Table 1. The 45 pre-GPU verification tests.

2. Full per-transform accuracy (5-seed mean)

Arm	I	R	R2	R3	H	RH	R2H	R3H
Original SFT	0.999	1.000	1.000	1.000	1.000	0.999	0.999	0.999
Augmentation	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Output cons.	1.000	0.999	1.000	0.999	0.999	0.999	0.999	0.998
Latent inv.	0.999	1.000	1.000	1.000	1.000	1.000	1.000	1.000
EquiOrient	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Wrong geom.	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

Table 2. Per-transform accuracy on test set (5-seed mean). Held-out elements: R2, R3, RH, R2H, R3H. All arms at ceiling.

3. Training loss curves (per epoch)

4. Individual seed results

5. Identifiability audit

The collision matrix compares $\rho(g)$ vs. $\tilde{\rho}(g)$ for all 8 group elements:

WACV #*****

Arm	ans ep1	ans ep2	struct ep1	struct ep2
Augmentation	3.217	0.019	—	—
Output cons.	5.109	0.986	0.686	0.459
Latent inv.	4.337	0.606	0.303	0.222
EquiOrient	4.106	0.211	0.318	0.108
Wrong geom.	4.424	0.388	0.351	0.174

Table 3. Per-epoch answer loss and structural loss (seed 101, N=512). Structural losses are nonzero and convergent (manipulation check passed).

Arm	s101	s202	s303	s404	s505
Original SFT	0.9984	1.0000	0.9988	0.9998	1.0000
Augmentation	1.0000	1.0000	1.0000	1.0000	1.0000
Output cons.	0.9963	0.9996	1.0000	1.0000	1.0000
Latent inv.	0.9990	1.0000	1.0000	1.0000	1.0000
EquiOrient	1.0000	1.0000	1.0000	1.0000	1.0000
Wrong geom.	1.0000	1.0000	1.0000	1.0000	1.0000

Table 4. Unseen-group accuracy per seed.

Element	$\rho(g) = \tilde{\rho}(g)$?	Distinguishes?
I	yes (trivially)	no
R	no ($R_{90} \neq -I$)	yes
R^2	no ($-I \neq I$)	yes
R^3	no ($R_{270} \neq -I$)	yes
H	yes (both $= H$)	no
RH	no ($R_{90}H \neq -IH$)	yes
R^2H	no ($-H \neq H$)	yes
R^3H	no ($R_{270}H \neq -IH$)	yes

Table 5. Collision matrix. The 5 held-out elements all distinguish correct from wrong representations. No collisions on the unseen set.

6. Bootstrap analysis

Hierarchical paired bootstrap: resample training seeds, then test scene IDs within each seed, preserving method pairing.

- Mean ΔA (EquiOrient − Augmentation): 0.0000 (at N=512, all ceiling)
- 95% CI: $[0.0000, 0.0000]$
- Individual seed deltas: all exactly 0.0000 at $N \geq 512$
- Minimum meaningful effect: 0.03 (3pp)
- Conclusion: no evidence of meaningful behavioral benefit at $N \geq 512$

7. Clean N=128 deterministic results (15 matched seeds)

Under the fully deterministic protocol (seeds fixed before model initialization; SHA-256-seeded data noise), we trained four key arms at N=128 with fifteen matched seeds.

Predeclared paired contrasts:

- EquiOrient − Augmentation: −0.93pp, 95% CI $[-2.28, +0.46]$pp, paired t-test $p = 0.215$, wins 5/15 seeds. Not significant.
- EquiOrient − Wrong geometry: +0.68pp, 95% CI $[+0.49, +0.89]$pp, paired t-test $p < 0.0001$, wins 15/15 seeds. Significant: behavior distinguishes the correct law from the incorrect one.

At N=128 the low-data regime does not ceiling: correct structural supervision does not outperform uncon-

Seed	Aug.	Eq.	Wrong	Eq−Aug (pp)
101	0.9846	0.9576	0.9494	−2.70
202	0.9139	0.9549	0.9510	+4.10
303	0.9516	0.9471	0.9430	−0.45
404	0.9867	0.9439	0.9373	−4.28
505	0.9297	0.9428	0.9338	+1.31
606	0.9492	0.8945	0.8891	−5.47
707	0.9682	0.9361	0.9219	−3.21
808	0.9689	0.9457	0.9436	−2.32
909	0.9699	0.9500	0.9424	−1.99
1010	0.9303	0.9404	0.9258	+1.01
1111	0.9703	0.9637	0.9596	−0.66
1212	0.9658	0.9516	0.9428	−1.42
1313	0.9289	0.9582	0.9576	+2.93
1414	0.9205	0.9437	0.9408	+2.32
1515	0.9826	0.9512	0.9408	−3.14
Mean	0.9547	0.9454	0.9386	−0.93

Table 6. Per-seed unseen-group accuracy at N=128, 15 matched deterministic seeds.

strained augmentation, but behavior is reliably sensitive to whether the imposed geometric law is correct. At $N \geq 512$ all of these contrasts vanish at ceiling.

8. Dataset statistics

Property	Value
Scenes (total)	4,096
Scenes (train pool)	2,048
Scenes (dev)	512
Scenes (val)	512
Scenes (test)	1,024
Examples per scene	8 (all D4 elements)
Total examples	32,768
Target size range	2–4 px
Distractors per scene	12–20
Color palette	3 muted hues
Background	(155, 152, 148)
Per-pixel noise	amplitude 12
Canvas size	192×192 px
Sparse exposure	identity + 1 generator (50/50 H/R)

Table 7. v4 dataset parameters.

9. Compute ledger

10. Reproducibility

- Code: All code on branch research/equiorient-phase2 at commit 9af8d25.
- Results: 30 JSON files in results/phase2_confirmatory/, one per (arm, seed) combination.
- Frozen ledger: equiorient/freezes/phase2_confirmatory.yaml committed before inspecting confirmatory results.
- Modal scaffold: modal/equiorient_phase2.py (serverless, pinned stack: torch 2.8.0+cu128, transformers==4.57.6, Qwen3-VL-8B @ 0c351dd0).
- Verification: 45-test pre-GPU gate (equiorient/tests/test_phase2_gate.py) must pass before any GPU run.

Run	Platform	Outcome	GPU time
v1 dev (2 jobs)	L40S (Modal)	passed, ceiling	∼34 min
v3 dev (2 jobs)	L40S (Modal)	passed, ceiling	∼34 min
v4 dev (2 jobs)	L40S (Modal)	passed, ceiling	∼34 min
Confirmatory (30)	L40S (Modal)	VALID	∼8.5 h
Total			∼9.5 h

Table 8. All GPU runs. Dev runs established the ceiling; confirmatory runs collected the final results.